SERVIDOR DHCP

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Um servidor DHCP é responsável pela atribuição de IPs para dispositivos em uma rede.

Passo a passo para a configuração de um servidor DHCP

Passo 1: configurar um ip estático para a máquina virtual.

```
adrian@adrian-VirtualBox:~$ sudo nano /etc/netplan/01-netcfg.yaml
```

```
  GNU nano 7.2                    /etc/netplan/01-netcfg.yaml
network:
  version: 2
  renderer: networkd
  ethernets:
    enp0s3:
      dhcp4: no
      macaddress: 08:00:27:EB:AA:CC
      addresses: [192.168.10.1/24]
```

Passo 2: aplicar as configurações de rede.

```
adrian@adrian-VirtualBox:~$ sudo netplan apply

** (generate:6275): WARNING **: 09:58:44.831: Permissions for /etc/netplan/01-netcfg.yaml are too open. Netplan configur
ation should NOT be accessible by others.

** (generate:6275): WARNING **: 09:58:44.831: Permissions for /etc/netplan/01-network-manager-all.yaml are too open. Net
plan configuration should NOT be accessible by others.

** (process:6273): WARNING **: 09:58:45.638: Permissions for /etc/netplan/01-netcfg.yaml are too open. Netplan configura
tion should NOT be accessible by others.

** (process:6273): WARNING **: 09:58:45.638: Permissions for /etc/netplan/01-network-manager-all.yaml are too open. Netp
lan configuration should NOT be accessible by others.

** (process:6273): WARNING **: 09:58:45.719: Permissions for /etc/netplan/01-netcfg.yaml are too open. Netplan configura
tion should NOT be accessible by others.

** (process:6273): WARNING **: 09:58:45.719: Permissions for /etc/netplan/01-network-manager-all.yaml are too open. Netp
lan configuration should NOT be accessible by others.
```

Passo 3: conferir se o ip foi definido.

```
adrian@adrian-VirtualBox:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:eb:aa:cc brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.1/24 brd 192.168.10.255 scope global noprefixroute enp0s3
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:feeb:aacc/64 scope link
       valid_lft forever preferred_lft forever
```

Passo 4: instalar o pacote de servidor dhcp.

```
adrian@adrian-VirtualBox:~$ sudo apt install isc-dhcp-server
Lendo listas de pacotes... Pronto
Construindo árvore de dependências... Pronto
Lendo informação de estado... Pronto
isc-dhcp-server já é a versão mais recente (4.4.3-P1-4ubuntu2).
0 pacotes atualizados, 0 pacotes novos instalados, 0 a serem removidos e 189 não atualizados.
```

Passo 5: editar arquivo padrão do dhcp.

```
adrian@adrian-VirtualBox:~$ sudo nano /etc/default/isc-dhcp-server
```

Passo 6: definir interface de rede.

```
INTERFACESv4="enp0s3"
INTERFACESv6=""
```

Passo 7: definir configuração de rede dhcp.

```
adrian@adrian-VirtualBox:~$ sudo nano /etc/dhcp/dhcpd.conf
```

```
subnet 192.168.10.0 netmask 255.255.255.0 {
        range 192.168.10.100 192.168.10.200;
        option routers 192.168.10.1;
        option subnest-mask 255.255.255.0;
        option domain-name-servers 8.8.8.8, 8.8.4.4;
        default-lease-time 600;
        max-lease-time 7200;
}
```

Passo 8: Iniciar o servidor.

```
adrian@adrian-VirtualBox:~$ sudo systemctl start isc-dhcp-server
adrian@adrian-VirtualBox:~$ sudo systemctl enable isc-dhcp-server
```

Passo 9: Conferir status.

```
adrian@adrian-VirtualBox:~$ sudo systemctl status isc-dhcp-server
● isc-dhcp-server.service - ISC DHCP IPv4 server
     Loaded: loaded (/usr/lib/systemd/system/isc-dhcp-server.service; enabled; preset: enabled)
     Active: active (running) since Mon 2025-09-29 10:45:59 -03; 8s ago
       Docs: man:dhcpd(8)
   Main PID: 11082 (dhcpd)
      Tasks: 1 (limit: 4603)
     Memory: 3.8M (peak: 4.3M)
        CPU: 12ms
     CGroup: /system.slice/isc-dhcp-server.service
             └─11082 dhcpd -user dhcpd -group dhcpd -f -4 -pf /run/dhcp-server/dhcpd.pid -cf /etc/dhcp/dhcpd.conf enp0s3

set 29 10:45:59 adrian-VirtualBox sh[11082]: Sending on   LPF/enp0s3/08:00:27:39:d0:57/192.168.10.0/24
set 29 10:45:59 adrian-VirtualBox sh[11082]: Sending on   Socket/fallback/fallback-net
set 29 10:45:59 adrian-VirtualBox dhcpd[11082]: Sending on   LPF/enp0s3/08:00:27:39:d0:57/192.168.10.0/24
set 29 10:45:59 adrian-VirtualBox dhcpd[11082]: Sending on   Socket/fallback/fallback-net
set 29 10:45:59 adrian-VirtualBox dhcpd[11082]: Server starting service.
set 29 10:46:01 adrian-VirtualBox dhcpd[11082]: DHCPREQUEST for 10.0.2.15 from 08:00:27:39:d0:57 via enp0s3: ignored (n>
set 29 10:46:03 adrian-VirtualBox dhcpd[11082]: DHCPDISCOVER from 08:00:27:39:d0:57 via enp0s3
set 29 10:46:04 adrian-VirtualBox dhcpd[11082]: DHCPOFFER on 192.168.10.102 to 08:00:27:39:d0:57 (adrian-VirtualBox) vi>
set 29 10:46:06 adrian-VirtualBox dhcpd[11082]: DHCPDISCOVER from 08:00:27:39:d0:57 (adrian-VirtualBox) via enp0s3
set 29 10:46:06 adrian-VirtualBox dhcpd[11082]: DHCPOFFER on 192.168.10.102 to 08:00:27:39:d0:57 (adrian-VirtualBox) vi>
lines 1-21/21 (END)
```

Passo 10: configurar a rede da máquina virtual com rede interna, criar outra máquina virtual (pode ser em modo teste) e também colocá-la como rede interna. Conferir se recebeu um ip no intervalo definido anteriormente.

```
ubuntu@ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gr
oup default qlen 1000
    link/ether 08:00:27:26:20:65 brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.101/24 brd 192.168.10.255 scope global dynamic noprefixroute
 enp0s3
```

Conclusão

Utilizando de um servidor DHCP, é possível, de forma dinâmica, atribuir endereços IP para diversos dispositivos dentro de uma rede. Pode ser amplamente configurado de acordo com as necessidades e preferências de cada um.